

Executive Summary

Predicting Severe Injury in European Football, 2020–2025

WAI Project · Sports Analytics · Akshay Thomas, IIM Ranchi

The question

Can a model predict, at the moment an injury is assessed, whether it will keep a player out for 28 days or more? A severe injury costs a club a squad place, a wage, and often a season; knowing early which injuries will become severe would change how a medical department triages and how a manager plans.

The dataset is 15,603 recorded injuries to 4,081 players across five European leagues over five seasons, enriched with Transfermarkt player profiles and career injury histories.

The headline result

Metric	Value	Reading
Best model	XGBoost	max_depth 3, learning rate 0.05
Test ROC-AUC	0.747	Meaningfully better than chance (0.500)
Recall on severe injuries	0.704	Catches 7 of every 10 severe cases
Precision on severe injuries	0.526	Nearly half of all flags are false alarms
Test set	3,906 injuries	1,021 players, none seen in training

The three models tested — logistic regression, random forest and XGBoost — finish within 0.006 AUC of one another. No model is meaningfully better than the others on this data, which is itself a finding: the signal available is largely accessible to a simple linear model, and the flexibility of tree ensembles buys almost nothing.

The principal contribution: a target leak found after modelling

The most consequential work in this project was not building the models but auditing them. An initial run produced an AUC of 0.801, led by a feature describing the player's average injury duration. The size of that feature's lead prompted a check of how it was constructed.

Five features summarised each player's injury history, built by aggregating their entire Transfermarkt record. That record spans 1973–2025 and therefore contains the very injuries being predicted. Half of all injury records in the file fall inside the modelling window, and for 54% of players the whole recorded history sits inside it; the median player has three career injuries. Each aggregate was, in effect, averaging in its own answer.

The features were rebuilt "as-of" each injury date, admitting only injuries that began strictly earlier. All three models were then refitted on the identical split, grids and random seed.

Model	AUC (leaked)	AUC (corrected)	Change
XGBoost	0.8012	0.7472	–0.0539
Random Forest	0.7921	0.7435	–0.0486
Logistic Regression	0.7758	0.7417	–0.0341

Between two-thirds and four-fifths of the apparent predictive signal in every history feature was the target leaking back in. The reported 0.747 is the honest figure; the 0.801 is the finding.

What actually predicts severity

Correcting the leak changed the answer, not just the score. Before correction the model was led by average injury duration. After correction it is led by injury type: illness, the residual "other" category, and knee injuries account for most of the model's behaviour, while the strongest history feature falls to sixth.

This is corroborated independently by the Phase 3 chi-square tests, where injury category scores 2,165.6 against severity — an order of magnitude above league (235.7) or position (24.7). The defensible conclusion is that what kind of injury a player sustains predicts its severity; who the player is barely does.

A secondary finding runs against intuition: players with longer injury records suffer slightly less severe injuries (median 10 prior injuries for severe cases against 12 for non-severe). Durable players who play often accumulate many minor knocks, so a long record signals availability rather than fragility.

Method and its constraints

- Observations are not independent. 15,603 injuries belong to 4,081 players, with up to 21 injuries for a single player. The train/test split and all cross-validation are grouped by player, so no individual appears on both sides; without this, repeated players inflate every score.
- Class imbalance is mild at 1.8:1, so class weighting was used rather than SMOTE. Synthesising minority rows at this ratio adds distortion without adding information.
- Leakage was audited twice — before modelling (four features removed) and again after (five features rebuilt). The second audit found the larger problem, which is the central methodological lesson of the project.
- Logistic regression violates the linearity-of-the-logit assumption for height and player age. This was predicted to place it last, and it does, but by a margin too small to treat as a confirmed prediction.

Limitations

- No exposure data. Minutes played, training load, match congestion and travel are unavailable, and are among the strongest known injury predictors in the sports-science literature. This caps what any model on this data can achieve.
- Only recorded injuries appear. Knocks managed without a reported absence are absent, so the model learns the severity distribution of reported injuries, not of all injuries.
- One feature still looks ahead. A count of each player's injuries across the whole window remains in the model and should be made as-of or dropped.
- Threshold tuning predates the correction and needs repeating against the corrected model, ideally on a validation fold rather than the test set.
- The 28-day threshold is a modelling choice following clinical convention, not a property of the data. Every relationship reported is associational, not causal.

Conclusion

A leakage-free model identifies roughly seven in ten severe injuries at assessment, at the cost of nearly one false alarm for every correct flag. For a medical department weighing a missed severe injury against an unnecessary precaution, that trade-off is favourable, and the model is best used to support triage by diagnosis rather than to profile players by injury record.

The wider lesson is methodological. An initial result of 0.801 looked like the better outcome and would have been straightforward to report. It was wrong, and the evidence that it was wrong was visible in the feature importances before any external check. The value of this project lies less in the 0.747 than in the audit that produced it.

Full analysis in `WAI_Soccer_Injury_Prediction.ipynb` (Phases 0–5b). Assumptions, diagnostics and the leakage repair are documented in the accompanying Phases 4 & 5b report.